

Stat 512 Final Project

Project Topic: Exploring Factors Influencing Highway MPG to Combat Climate Change

Group number: 7

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Project YouTube link: <https://youtu.be/JAS7k1EFxW4>

Project Background Introduction: The automobile industry significantly contributes to greenhouse gas emissions, and fuel efficiency metrics like highway miles per gallon (MPG) are important metrics for reducing fossil fuel consumption and producing a more climate-conscious vehicle. Understanding the factors that influence highway MPG, such as vehicle weight, fuel systems, and engine types, can guide sustainable vehicle design and policy development to mitigate climate change.

Current research highlights vehicle weight, fuel systems, and engine types as key determinants of fuel efficiency. Studies show that lighter vehicles and certain fuel systems (e.g., port fuel injection) are more efficient. Diesel engines outperform gasoline engines, and engine designs such as inline engines are generally more efficient than those used in sports cars.

Major Research Question: What factors in the car manufacturing process most effectively predict highway MPG, and how can understanding these relationships help address climate change?

Exploratory Research Questions:

1. Which MPG has a greater effect on vehicle price, city MPG or Highway MPG?
2. Does the fuel and engine type affect the Highway MPG?
3. Does the impact of curb weight on Highway MPG vary depending on Engine Type?
4. Does curb weight and engine size significantly impact horsepower of a vehicle?

Project Result Highlight:

1. Impact of City and Highway MPG on Vehicle Price: The analysis reveals a significant interaction between city MPG and highway MPG in determining vehicle price, supporting the idea that fuel efficiency plays a role in the cost of a vehicle. This highlights the importance of factoring both city and highway fuel economy into vehicle pricing, which could guide policies and consumer decisions toward more sustainable transportation.
2. Impact of Curb Weight and Engine Size on Horsepower: Both curb weight and engine size have a significant effect on vehicle horsepower, with engine size having the more notable impact. This reinforces the need to carefully consider these factors in vehicle design, as optimizing them can help improve both fuel efficiency and sustainability in manufacturing.
3. Curb Weight's Negative Impact on Highway MPG and Engine Type Moderation: While curb weight negatively impacts highway MPG, engine type plays a key role in reducing this effect. The OHCF engine type, in particular, has the most significant positive impact on highway MPG, suggesting that choosing the right engine type can enhance fuel efficiency, which is critical for lowering emissions in the transportation sector.

Project data introduction: The dataset, "Automobile," is from the UC Irvine Machine Learning Repository and includes observations from the 1985 *Ward's Automotive Yearbook*. This study

examined 205 different vehicles and collected data on 26 parameters, in which we focused on the following 8:

Variable	Type	Description
Fuel Type	Binary Categorical	Diesel or Gasoline
Engine Type	Nominal Categorical	Types: DOHC, DOHCV, L, OHC, OHCF, OHCV, ROTOR
Curb Weight	Ratio Continuous	Vehicle weight without cargo (pounds)
Engine Size	Ratio Continuous	Displacement volume of the engine (liters)
Horsepower	Ratio Continuous	Engine output in horsepower
City MPG	Ratio Continuous	Miles per gallon in city driving conditions
Highway MPG	Ratio Continuous	Miles per gallon in highway driving conditions
Price	Ratio Continuous	Price of the vehicle in dollars

What We Learned & Most Challenging Portion: The portion of the analysis that was particularly challenging was the need to address non-normality and heteroscedasticity in the residuals, especially when outliers were present, which contributed to distorted results. Additionally, ensuring the model accurately captured complex interactions, like those between categorical and continuous variables, required careful attention. These interactions added complexity, as they involved non-linear relationships that needed to be modeled correctly to avoid misinterpretation and ensure the model's validity.

Future Advice: Pick your data and research questions based on a topic you know, as it will help you logically think through the questions and understand the data, which we learned when we switched to a more familiar dataset.

Background & Overview

Background: The transportation sector is a major contributor to greenhouse gas emissions, with fuel consumption in vehicles playing a significant role. Highway miles per gallon (MPG) is a critical metric for understanding fuel efficiency, and improving this efficiency directly reduces fossil fuel consumption and associated emissions. Investigating the factors that influence highway MPG, such as vehicle weight, fuel systems, and engine types, provides valuable insights into designing more efficient vehicles. By identifying these factors, we can guide manufacturing decisions and policy measures that support sustainable transportation, contributing to global efforts to mitigate climate change and reduce our carbon footprint.

Literature Review: Highway miles per gallon (MPG) serves as a critical metric for understanding vehicle fuel efficiency, a key driver in addressing climate change through reduced greenhouse gas emissions. Identifying the factors that influence highway MPG is essential for optimizing automobile design and manufacturing processes. This literature review synthesizes existing research

on these factors, focusing on findings that connect to our research questions and variables of interest in our dataset.

Vehicle weight is a well-documented determinant of fuel efficiency. For instance in her Ph.D Thesis for MIT Cheah, Lynette found that a “10% reduction in vehicle weight can cut fuel consumption by about 7%” (Cheah). This means in our model we would expect vehicle weight to be negatively correlated with Highway miles per gallon since that is a measure of fuel efficiency.

There are eight types of fuel systems in our dataset: 1bbl (1-barrel carburetor), 2bbl (2-barrel carburetor), 4bbl (4-barrel carburetor), IDI (Indirect Injection), MFI (Multi-Port Fuel Injection), MPFI (Multi-Point Fuel Injection), SPDI (Single-Port Direct Injection), and SPFI (Single-Port Fuel Injection). Each system differs in how it delivers fuel and impacts engine performance and efficiency.

The type of fuel system in a car can influence its performance and fuel efficiency. A study by Seoksu Moon found that "the DI [Direct Injection system] decreased engine thermal efficiency in high-load conditions, even when using the same injection timing as PFI [Port Fuel Injection systems] (Moon)." In our dataset, MPFI and SPFI are categorized as PFI systems, while SPDI (Single-Port Direct Injection) falls under Direct Injection systems. High-load conditions, such as those involving high engine pressure, commonly occur during highway driving. Based on this, it's likely that MPFI and SPFI systems will have slightly better highway MPG than SPDI systems. However, the study suggests that the difference in efficiency is small, so it might not significantly affect predictions in our model.

For carburetor types—1bbl (1-barrel), 2bbl (2-barrel), and 4bbl (4-barrel)—the 1bbl carburetor is generally the most fuel-efficient, followed by the 2bbl, with the 4bbl being the least efficient. This is because engines with more barrels can deliver more power but also consume more fuel. As a result, engines with higher-barrel carburetors are expected to have lower highway MPG.

The type of fuel used also plays a major role in fuel efficiency. According to the U.S. Department of Energy, “Diesel engines are more powerful and 30%-35% more efficient than similar-sized gasoline engines.” This means that, on average, diesel-powered vehicles should have higher highway MPG compared to gasoline-powered vehicles (US Department of Energy).

The engine types in our data are DOHC (Dual Overhead Camshaft), DOHCV (Dual Overhead Camshaft, V-Configuration), I (Inline Engine), OHC (Overhead Camshaft), OHCF (Overhead Camshaft, Flat Engine), OHCV (Overhead Camshaft, V-Configuration), and Rotor (Rotary Engine). The Dual Overhead engines are most common in sports cars which tend to be less energy efficient. Inline Engines are most common in small, compact cars so it's expected that these engines will be more fuel efficient than say a Dual Overhead engine. In our dataset the Rotor engines are only by one car: the Mazda Hatchback, which tends to be fuel inefficient. Lastly, the Overhead Camshaft engines are the by far the most common type of engine and is used in a variety of cars, so it's unclear how impactful it will be on energy efficiency.

This review identifies key factors influencing highway miles per gallon (MPG), specifically vehicle weight, fuel and fuel system type, and engine type. Studies show a clear negative correlation between vehicle weight and fuel efficiency, while port fuel injection systems like MPFI and SPFI typically outperform direct injection systems like SPDI under highway driving conditions. Fuel type

also matters significantly, with diesel engines being more fuel-efficient than gasoline engines. Carburetor systems exhibit a predictable trend, where fewer barrels correlate with higher MPG, and engine types vary in efficiency based on design and application; for example, inline engines, common in compact cars, are generally more efficient than dual overhead camshaft (DOHC) engines found in sports cars. These findings provide a solid foundation for predicting highway MPG within the dataset, offering insights into how vehicle design and fuel choices impact energy efficiency and sustainability.

Research Questions and Hypothesis

Nature of Study: This research project utilizes observational data titled “Automobile” from the UC Irvine Machine Learning Repository, originally published in the 1985 edition of *Ward’s Automotive Yearbook*. This study examined 205 different vehicles and collected data on the following parameters: Make, Fuel Type, Aspiration, Number of Doors, Body Style, Drive Wheels, Engine Location, Wheel Base, Length, Width, Height, Curb Weight, Engine Type, Number of Cylinders, Engine Size, Fuel System, Bore, Stroke, Compression Ratio, Horsepower, Peak RPM, City MPG, Highway MPG, Price, Average Loss per Car per Year (Normalized Losses), and Risk Factor Symbol.

Variable Explanation: The variables of interest this project will focus on include Fuel Type, Curb Weight, Engine Type, Engine Size, Horsepower, City MPG, Highway MPG, and Price. Of these variables, Fuel Type is a binary categorical variable, Engine Type is a nominal categorical variable, and Curb Weight, Engine Size, Horsepower, City MPG, Highway MPG, and Price are ratio continuous variables. Of the 205 observations, 46 had blank values and were removed from the data set to allow for computing. Following data cleaning, we have 159 remaining observations to base the following models off of.

Limitations of Prediction Power: Since the Automobile dataset is from 1985, there is potential that the data is outdated, reflecting automotive trends and technology from the 1980s, and does not include modern features like electric vehicles or advanced driver-assistance systems, making its relationships potentially irrelevant today. The dataset also suffers from sampling bias, as it only includes cars from Ward’s Automotive Yearbook, which may overrepresent certain brands or car types. Missing important variables like emissions and maintenance costs limits its predictive accuracy, and missing data for attributes like horsepower can introduce bias. With only 205 records, the smaller sample size makes the dataset prone to overfitting and limits its generalizability.

Overarching Research Question: What factors in the car manufacturing process most effectively predict highway MPG, and how can understanding these relationships help address climate change?

Research Question 1 - Nate Nolan

Research Question: Is there an interaction effect between city MPG and highway MPG on vehicle price?

Tie to Overarching Research Question

Generally speaking, a vehicle with more efficient fuel usage is going to be better for the environment than a vehicle with bad fuel efficiency simply due to carbon emissions. This goes for both city fuel efficiency as well as highway fuel efficiency. Identifying whether these indicators of an environmentally-friendly vehicle interact with one another to affect vehicle price could be useful in grasping whether or not environmentally-friendly vehicles are being made affordable. An interaction effect existing would suggest that the two different types of fuel efficiency have a unique relationship that allows them to together either drive down vehicle prices or drive up vehicle prices. Making sure vehicles that are more fuel efficient are affordable is an important step in enabling a majority of vehicles on the road to be less environmentally harmful.

Null Hypothesis: $H_o : \beta_{City\ MPG * Highway\ MPG} = 0$

There is no interaction effect between city MPG and highway MPG in relation to the price of the vehicle. This suggests that these two indicators of climate friendliness together do not interact to increase or decrease vehicle price. These are only indicators of vehicle price separately.

Alternative Hypothesis: $H_a : \beta_{City\ MPG * Highway\ MPG} \neq 0$

There is an interaction effect between city MPG and highway MPG in relation to the price of the vehicle. This suggests that these two indicators of climate friendliness together do interact to increase or decrease vehicle price, making climate-friendly vehicles either more or less affordable.

Methodology

- I. **Model Building:** To test the hypotheses and answer the research question, a multiple linear regression model was developed. The dependent variable is vehicle price, city MPG and highway MPG as predictors, along with their interaction term. In-depth model limitations are discussed earlier in the paper, but major issues with this test include outdated data and missing values decreasing the sample size, potentially resulting in overfitting of the data. Initially, the model assumes ordinary least squares (OLS) regression:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2) + \epsilon$$

Where:

- Y: Vehicle Price (Continuous)
- X_1 : City MPG (Continuous)
- X_2 : Highway MPG (Continuous)
- β_0 : Intercept (Baseline vehicle price when predictors are 0)
- β_1 : Coefficient for city MPG
- β_2 : Coefficient for highway MPG
- β_3 : Interaction coefficient for city MPG and highway MPG
- ϵ : Error term

II. Diagnostics

Residuals vs. Predictor Variable Plot: This plot appears to show heteroskedasticity as lower X values appear to be much more variable than higher X values.

Residuals vs. Fitted Values Plot: This plot shows that there are some potential outliers towards the center of the dataset as the center has the most observations the furthest from the regression line.

Normal Q-Q Plot: This plot shows that towards the upper end of the dataset observations begin to become much more variable as they distance themselves further from the regression line.

Shapiro-Wilk Test: With a p-value of 1.075×10^{-8} it can be concluded that the initial model does not follow a normal distribution.

Breusch-Pagan Test: With a p-value of 2.615×10^{-7} it can be concluded that the initial model does contain heteroskedasticity.

Cook's Distance: Thirty points were identified as being outliers using Cook's distance. This suggests that these data points are influencing the distribution to make it non-normal and should be dropped from the model if possible.

III. Remedial Procedures: After diagnostic testing, the assumptions that were violated include Non-Normality of Residuals as well as a lack of constant variance. The main indicator of heteroskedasticity was the initial Breusch-Pagan p-value of 9.445×10^{-9} . The main indicator of nonnormality was the Shapiro-Wilks test p-value of 2.316×10^{-8} . These issues were addressed by the following transformations of the model:

- I. Outliers identifiers based on Cook's distance were removed.
- II. Box-Cox transformation on Y to address heteroskedasticity. The ideal λ was found to be -0.5051.
- III. Logarithmic transformation on X_1 and X_2 to address nonnormality

Final Model: The Breusch Pagan results after the model had been transformed suggest that the transformation did adequately address the violation of heteroskedasticity with a p-value of 0.265. The new Shapiro-Wilks test didn't reach the ideal threshold to make the conclusion of normality, but the transformation did bring the model closer to normal at a p-value of 0.00103. After transformation, the new model looks like the following:

$$Y^\lambda \sim \beta_0 + \beta_1 \log(X_1) + \beta_2 \log(X_2) + \beta_3 \log(X_1) \log(X_2)$$

Results

- I. Hypothesis Discussion:** The analysis confirms that there is a visible interaction effect between city MPG and highway MPG on vehicle price, leading us to reject the null hypothesis. This interaction effect is demonstrated by the extremely low p-value for β_3 of 0.02399 which is lower than our rejection level of $\alpha = 0.05$. This suggests that the combined influence of these predictors significantly contributes to explaining variations in vehicle price.
- II. Conclusion:** In summary, while city MPG and highway MPG each have a positive impact on vehicle price, the combined influence of these predictors has an estimated negative impact of -0.002594. This supports the alternative hypothesis (H_a), highlighting that there is an interaction effect between city MPG and highway MPG on vehicle price.

The transformed model, which contains a Box-Cox transformation on the dependent variable of vehicle price and a logarithmic transformation on the two predictor variables of city MPG and highway MPG was found to have the greatest predictive power out of all attempted transformations with its lack of heteroskedasticity as well as its closeness to normality. For future implementations of a similar model, it is recommended that a closer to normal distribution for highway and city MPG's are found to make the model more accurate. As of now the model is only close to normal, not normal according to the Shapiro-Wilks test, but the transformed model presented is as close to normal as we got.

Research Question 2 - Ace Setiawan

Research Question: How do curb weight and engine size impact the horsepower produced by an automobile?

By examining how these factors influence horsepower, we gain a better understanding of trade-offs in vehicle design. Optimizing these factors can lead to more fuel-efficient vehicles with adequate performance, thus improving highway MPG and reducing environmental impact.

Tie to Overarching Research Question: By understanding how curb weight and engine size influence horsepower in an automobile, we gain a better understanding of the trade-offs in vehicle design. Optimizing these factors can lead to more fuel-efficient vehicles with adequate performance that meets the needs of customers, thus improving highway MPG and reducing the environmental impact of gasoline powered vehicles.

Null Hypothesis: $H_0 : \beta_{\text{curb-weight}} = 0 \text{ and } \beta_{\text{engine-size}} = 0$

The amount of horsepower an automobile produces is not significantly impacted by the curb weight or the engine size of a vehicle.

Alternative Hypothesis: $H_o : \beta_{\text{curb-weight}} \neq 0 \text{ or } \beta_{\text{engine-size}} \neq 0$

The amount of horsepower an automobile produces is significantly impacted by the curb weight and the engine size of a vehicle.

Methodology

- I. **Model Building:** To test the hypotheses as well as answering the research question of how does curb weight and engine size of a vehicle determine the horsepower it produces, we can build a multiple linear regression model. With horsepower being the dependent variable and curb weight (pounds) and engine size (liters) being the features of the model. Initially, the model assumes ordinary least squares (OLS) regression. However, after evaluating OLS assumptions, weighted least squares (WLS) is adopted as the final model due to the presence of heteroscedasticity within the initial model. The weights used for the WLS regression is $1/|\text{Residuals}|$.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

Where:

y : Horsepower

- β_0 : Intercept (baseline horsepower when features equal 0)
 β_1 : Coefficient for curb weight
 β_2 : Coefficient for engine size
 ϵ : Error term

II. Diagnostic & Remedial Procedures:

Breusch-Pagan Test: indicates no heteroscedasticity in the model (p-value = 0.7614)

Shapiro-Wilks Test: indicated a non-normal distribution (p-value = 1.587e-07)

K-fold cross validation: using 10 folds shows a model that has a good level of variance explained by the predictors (RMS = 17.13, R-squared = 0.6983)

Results

- I. Hypothesis Discussion:** From the summary of the model obtained using R (Reference VIII), we can gather the coefficient for our predictors, curb weight's estimated coefficient is 0.016681 while engine size's estimated coefficient is 0.597774. Meaning for every additional unit of curb weight in pounds in a vehicle, the horsepower of said vehicle increases by approximately 0.0167 when holding the engine size constant. Similarly, for every additional liter of engine size, the horsepower of a vehicle increases by approximately 0.598 when holding the curb weight constant.

Using the p-values provided for each coefficient in the model summary we can answer our research question of whether horsepower of a vehicle is impacted by curb weight as well as engine size. Both the p-value of curb weight and engine size in our model is very small and is significant at the 95% confidence level. We have enough evidence to reject the null hypotheses and can conclude that both curb weight and engine size has a significant impact on a vehicle's horsepower.

Our model as a whole is also concluded to be very significant looking at the F-statistic of 714.5 and p-value of 2.2e-16. Further solidifying our alternative hypotheses as both predictors explain a large portion of the variation in a vehicle's horsepower.

- II. Conclusion:** Based on the results of the hypothesis testing, both curb weight and engine size significantly impact horsepower. The overall WLS model is highly significant and the alternative hypothesis that the two coefficients are non-zero is accepted. Therefore, we conclude that these factors are crucial for determining an automobile's horsepower and can be an effective measure in environmentally conscious vehicle design moving forward.

Research Question 3 - Mollie Parker

Research Question: Does curb weight have the same impact on highway MPG across all engine types?

Tie to Overarching Research Question: Since the overarching research question explores which factors in the car manufacturing process most effectively predict highway MPG, my individual research question focuses on the interaction between curb weight and engine type and how they

relate to highway MPG. Specifically, I aim to determine: (1) whether curb weight impacts highway MPG differently across various engine types and (2) which engine type achieves the highest highway MPG, regardless of weight. By examining these relationships, I can identify engine types where curb weight has the least impact on highway MPG, helping me pinpoint the engine type that is most climate-conscious.

Null Hypothesis: $H_o : \beta_{curb\ weight}$ same for all engine types

The relationship between curb weight and highway MPG does not differ between the different Engine Types, which is used as a proxy for how climate-conscious a vehicle is, and is therefore consistent despite different models of engine.

Alternative Hypothesis: $H_a : \beta_{curb\ weight}$ differs for at least one of the engine types

The relationship between curb weight and highway MPG does differ between the different engine types, which is used as a proxy for how climate-conscious a vehicle is, and is therefore consistent despite different models of engine.

Methodology

- I. Model Building:** To test the hypotheses and answer the research question, a multiple linear regression model was developed. The dependent variable is highway MPG, with curb weight (pounds) and engine type as predictors, along with their interaction term. In-depth model limitations are discussed earlier in the paper, but major issues with this test include outdated data and missing values decreasing the sample size, potentially resulting in overfitting of the data.

It is also important to note that in this model, the engine type variable includes several categories, but due to missing data, two engine types, DOHCV and ROTOR, were not included in the final model. These engine types had insufficient data entries or contained NA values, which led to their exclusion from the analysis. As a result, the model uses the DOHC engine type as the baseline for comparison. Initially, the model assumes ordinary least squares (OLS) regression:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_{2L} + \beta_2 X_{2OHC} + \beta_2 X_{2OHCv} + \beta_2 X_{2OHCf} + \beta_3 X_1 X_{2L} + \beta_2 X_1 X_{2OHC} + \beta_2 X_1 X_{2OHCv} + \beta_2 X_1 X_{2OHCf} + \epsilon$$

Where:

- y : Highway MPG
- β_0 : Intercept (baseline highway MPG when features are at baseline: Engine Type DOHC)
- β_1 : Coefficient for curb weight
- β_2 : Coefficient for engine type
- β_3 : Interaction coefficient for curb weight and engine type
- ϵ : Error term

II. Diagnostic Tests

Residuals vs. Predictor Variable Plot: In the Residuals vs. Predictor Variable Plot, most data points cluster around the line, but a few have large residuals, both above the line and to the right. This suggests potential outliers and possible heteroscedasticity, where residual variance increases with larger predictor values.

Residuals vs. Fitted Values Plot: There are strong clusters around the line, with a concentration to the right. Additionally, numerous outliers are present, both above and below the line, indicating potential heteroscedasticity and non-linearity in the model.

Normal Q-Q Plot: The points between -1 and 1 are close to the line, suggesting normality in that range. However, outside this range, the points curve downward below -1 and upward above 1, indicating non-normality and potential outliers or skewness in the residuals.

Histogram of Residuals: The histogram shows a right skew, with points lying out to the right of the distribution. This suggests some departure from normality, but the central part of the distribution appears approximately normal, indicating that the residuals are mostly symmetrically distributed around the mean.

Boxplot for Outlier Detection: The boxplot reveals a higher concentration of outliers on the upper end, indicating that the model may struggle with larger residuals, particularly for higher values. This suggests potential issues with the model's fit for extreme data points.

Shapiro-Wilk Test: The p-value from the test was extremely small (7.195e-09), which indicates that the residuals deviate significantly from a normal distribution. Therefore, the assumption of normality for the residuals is violated.

Breusch-Pagan Test: The test result indicates that the p-value is 0.3641, which is greater than the commonly used significance level of 0.05. This suggests that there is no evidence of heteroscedasticity, meaning that the assumption of constant variance of residuals holds in the model.

Cook's Distance: No data points were identified as influential based on Cook's Distance (all values were below the threshold). This indicates that, overall, no single observation significantly impacts the model's results.

DFITS: The DFITS test did not identify any points as influential (no points exceeded the threshold), suggesting that the residuals are not overly affected by any individual data point in terms of leverage.

DFBETAS: The DFBetas test identified several influential points, including observations with indices 19, 31, 48, 69, 71, 108, 112, 113, 116, 117, among others. These points may have a disproportionate impact on the model coefficients and should be considered for further investigation.

VIF Values: The $GVIF^{(1/(2 \cdot Df))}$ values indicate adjusted measures of multicollinearity. Curb_weight has a value of 4.08, suggesting moderate multicollinearity, while Engine_type and Curb_weight:Engine_type have higher values of 8.88 and 8.95, respectively, indicating stronger multicollinearity.

III. Remedial Procedures: After diagnostic testing, the assumptions that were violated include Non-Normality of Residuals and Outliers. While Heteroscedasticity was initially a concern with the results of the Residuals vs. Predictor Variable Plot, the more advanced diagnostic

test, Breusch-Pagan, resulted in a p-value of greater than 0.05, thus we fail to reject the null hypothesis and accept that the residuals have constant variance. Thus, the main issues of concern are Non-Normality and Outliers. To address this the following measures were taken:

- I. Log Transformation to address Normality of Residuals - Figure IV
- II. Removal of Influential Points to address Outliers - Figure V

K-Fold Cross-Validation: To evaluate the performance of each remedial model, 10-fold cross-validation was employed, using 9 folds for training and 1 for testing iteratively. The RMSE values derived from cross-validation allow for a comparison of model accuracy, aiding in the selection of the best-performing model for predicting highway MPG.

- I. Log Transformation Model RMSE: 3.231
- II. Influential Points Removed Model RMSE: 3.908462

Final Model: The log-transformed model demonstrates superior predictive performance, achieving the lowest RMSE and accurately capturing the relationship between curb weight, engine type, and highway MPG. By addressing non-normality in the residuals, the log transformation ensured the model retained valuable information without oversimplifying the dataset. This process resulted in the final equation:

$$\text{Highway MPG} = \beta_0 + \beta_1 \cdot \log(\text{Curb Weight}) + \beta_2 \cdot \text{Engine Type} + \beta_3 \cdot (\log(\text{Curb Weight}) \times \text{Engine Type}) + \epsilon$$

Diagnostic improvements further support the model's reliability. After the transformation, the Breusch-Pagan test showed a p-value of 0.423, indicating no significant heteroscedasticity, and the Shapiro-Wilk test returned a p-value of 0.054, demonstrating that residuals were now closer to a normal distribution. Additionally, residual diagnostics, including Q-Q plots and histograms, confirmed reduced skewness and better alignment with normality assumptions.

To refine the model further, outliers were removed based on Cook's Distance, which stabilized residuals and improved the model fit. Categorical variables, particularly the levels associated with engine types, were carefully adjusted to ensure the robustness of the final model.

Results

- I. **Hypothesis Discussion:** From the summary of the log-transformed model, the estimated coefficients and p-values provide clear insights into the relationship between curb weight, engine type, and highway MPG. For DOHC engines, curb weight's negative effect on MPG is highly significant, with a coefficient suggesting that heavier vehicles experience a substantial reduction in fuel efficiency ($p < 2e-16$). Inline engines (Engine_typed) exhibit a lesser negative effect, with a coefficient of 5.68 and a p-value of 0.00201, indicating a moderate impact compared to DOHC engines. OHC engines show a marginally significant negative effect on MPG, with a p-value close to 0.085. In contrast, OHCF and OHCV engines demonstrate no significant relationship with MPG, suggesting these engine types are less influenced by variations in curb weight.

The p-values for the significant coefficients in the model are small and fall below the 95% confidence level threshold, providing sufficient evidence to reject the null hypothesis. This supports the conclusion that both curb weight and engine type significantly impact highway MPG. Furthermore, the interaction between curb weight and engine type emphasizes how certain engine designs can mitigate the negative effects of vehicle weight on fuel efficiency.

The overall model is highly reliable, as evidenced by its low RMSE and the diagnostic improvements achieved through the log transformation and outlier removal. This ensures that the coefficients accurately represent the underlying relationships, bolstering our confidence in the alternative hypothesis that curb weight and engine type interact meaningfully to affect highway MPG.

- II. Conclusion:** The log-transformed model provides the most accurate and statistically sound representation of the relationship between curb weight, engine type, and highway MPG. By addressing diagnostic challenges, including heteroscedasticity and non-normality, the model effectively resolved prior violations of assumptions. This analysis confirms that curb weight has a pronounced negative effect on highway MPG, with significant variations observed across engine types. DOHC engines demonstrated the strongest negative impact, while Inline engines showed a lesser effect, and other engine types exhibited weaker or no significant relationships.

The study underscores the importance of tailored diagnostic and remedial procedures in developing reliable predictive models. The findings highlight that, despite the inherent negative relationship between curb weight and fuel efficiency, some engine types can mitigate this effect, making them more favorable for achieving higher MPG. This analysis not only provides valuable insights for automotive design and fuel efficiency strategies but also demonstrates the utility of robust statistical modeling in uncovering nuanced relationships within complex datasets.

Research Question 4 - Daniel Armeanu

Research Question

Does the fuel, fuel system type or engine size affect the Highway MPG?

Tie to Overarching Research Question

This research question is relevant to combating climate change because fuel efficiency directly impacts greenhouse gas emissions. Higher MPG means less fuel consumption, reducing emissions like carbon dioxide. It's likely that the different fuels, fuel systems, and engine sizes could result in higher or lower efficiency making some combinations more effective in fighting climate change. By identifying the key factors influencing highway MPG, this research provides actionable insights for automakers and policymakers to design and regulate vehicles that minimize environmental harm while improving energy efficiency.

Null Hypotheses: $H_0: \beta_0 = \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_{1,2} = \beta_{1,3} = \beta_{1,4} = \beta_{1,5}$

Testing this hypothesis required some additional cleaning to the data. Firstly, there were eight fuel system types to test, but after removing observations with missing data points there were only six. Additionally, the fuel system type MFI had only one observation. Since MFI is usually synonymous with MPFI, this observation was recategorized as MPFI. Secondly, there were two fuel types: gasoline and diesel; however, all the diesel cars exclusively used the IDI fuel system while the gasoline cars exclusively used 1BBL, 2BBL, MPFI, and SPDI. This meant that the question about fuel type could be answered by examining fuel system type. This led to the following null hypothesis: The engine size or fuel system type has a significant impact on the highway MPG of a vehicle.

Alternative Hypotheses: At least one of the β 's excluding β_0 is not equal to 0.

Methodology

I. **Model Building:** To test the hypothesis a multiple linear regression model was developed.

The dependent variable is highway MPG and the independent variables are fuel system type and engine size. This original model had several model assumption violations making it not the ideal choice for the final model. Here is the original model:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_{1,2} X_1 X_2 + \beta_{1,3} X_1 X_3 + \beta_{1,4} X_1 X_4 + \beta_{1,5} X_1 X_5 + \epsilon$$

Where:

- y : Highway MPG
- β_0 : Intercept (baseline highway MPG with fuel system type 1BBL)
- β_1 : Coefficient for engine size
- X_1 : Engine Size
- X_2 : dummy variable that is 1 if the fuel system type is 2BBL and 0 otherwise
- X_3 : dummy variable that is 1 if the fuel system type is IDI and 0 otherwise
- X_4 : dummy variable that is 1 if the fuel system type is MPFI and 0 otherwise
- X_5 : dummy variable that is 1 if the fuel system type is SPDI and 0 otherwise
- $\beta_2, \beta_3, \beta_4, \beta_5$: intercept term when for X_2, X_3, X_4, X_5 respectively. This represents the change in baseline highway MPG for each fuel system when compared to 1BBL
- $\beta_{1,2}, \beta_{1,3}, \beta_{1,4}, \beta_{1,5}$: interaction terms between X_1 and X_2, X_3, X_4, X_5 respectively
- ϵ : normally distributed error term with constant variance

II. Diagnostic Tests:

Residuals vs. Predictor Variable Plot. In the Residuals vs Predictor Variable Plot most data points were clustered around the line, but it appears that the variance is higher for data points with low engine sizes. This suggests potential outliers and non-constant variance.

Residuals vs Fitted Values Plot. In the Residuals vs Fitted Value Plot most data points were clustered around the line, but it appears that the variance is higher for some fitted values than others. This suggests potential outliers and non-constant variance.

Normal Q-Q Plot. The point near the center of the graph cluster around the line however, the points where the tails are very far from the line indicating that the data could potential skew in the data and also non-normality.

Histogram of Residuals. The histogram shows a slight right skew since there are extreme positive values. Additionally there are more residuals concentrated near 0 than is usually expected if the residuals were normally distributed. This indicates non-normality.

Boxplot for Outlier Detection. The boxplot indicates several positive and negative outliers. These points may be influential and affect the model fit.

Shapiro-Wilk Test. The p-value from this test was very small ($2.118e-6$) indicating non-normality of the residuals. This agrees with the general observations from the Q-Q Plot, Histogram, and Boxplot.

Breusch-Pagan Test. The p-value from this test is not significant at 0.065, but it is relatively small when combined with the observations from the Residuals vs. Predictor Variable Plot and Residuals vs Fitted Value Plot could provide evidence for non-constant variance even if the test itself is not significant at a 0.05 level.

III. Remedial Procedures: After the diagnostic testing, it was clear that the normality assumption was violated and potentially the constant variance assumption. In addition, the Boxplot test indicated several potential outliers. To address these concerns a log transformation of y was done to address both non-constant variance and non-normality. After the transformation, Q-Q plot still indicated non-normality however the Shapiro-Wilks p-value increased to 0.00011. This indicates that the residuals are closer to a normal distribution than before. Additionally, Residuals vs Fitted Values Plot and the Residuals vs Engine Size appear closer to having constant variance. This was confirmed by a higher p-value of 0.32. The Boxplot and Histogram still indicated outliers although their number and magnitude seems to have been reduced. To combat the potential influence of outliers, a robust regression was performed on the transformed y values.

IV. K-fold Validation: A 10-fold cross validation was employed to compare whether the robust model was out-performing the transformed OLS model. The transformed OLS model ended up having a lower RMSE of 0.111 while the robust model had a RMSE of 0.1112. Since these two models have very comparable accuracy the K-fold Validation resulted in the transformed OLS model being used since it is simpler and more interpretable.

V. Final Model. The transformed OLS model was chosen as the final model since it had a comparable RMSE while being more interpretable. The final equation is given by

$$\log(y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_{1,2} X_1 X_2 + \beta_{1,3} X_1 X_3 + \beta_{1,4} X_1 X_4 + \beta_{1,5} X_1 X_5 + \epsilon$$

Additionally the model performed better on diagnostic measures such as Breusch-Pagan and Shapiro-Wilks making it a better model than the OLS.

Results: The model summary provided insights into the relationship between fuel system type, engine type and highway MPG. The most efficient fuel system was the IDI, which was the fuel system used exclusively by diesel vehicles. While the result was not statistically significant with a p-value 0.67, this result does agree with existing research that diesel vehicles are more efficient than

gasoline vehicles. Additionally the remaining results generally agree with exist research. For instance while not statistically significant the 2BBL engine did have a lower highway MPG than the 1BBL which is established in the literature. Also, it was expected that the SPDI engine would perform worse than the MPFI engine in highway conditions. This effect was also noticed and in fact the SPDI performed significantly worse than the 1BBL while the MPFI did not. Lastly, engine size was negatively correlated with highway MPG which is to be expected given a large engine will likely consume more fuel than smaller engines.

Conclusion: While several of the results did not have significant p-values, they did agree with the existing literature. As a result, we can conclude that the fuel system type does indeed have an effect on fuel efficiency. Additionally, diesel is a more efficient fuel source than gasoline and larger engines tend to be less efficient than smaller engines. This analysis could prove useful to car manufactures who are designing their cars to be more efficient to combat climate change.

Summary Discussion & Conclusion

Our analysis through examining 4 different research questions surrounding our general question reveals that multiple factors play a role in determining a vehicle's highway MPG and other vehicle attributes critical to efficiency and sustainability which is a point of interest for us. While the interaction between city MPG and highway MPG influences vehicle price, characteristics like engine size and curb weight have an equally important impact on its horsepower, a proxy for performance. Notably curb weight's effect on highway MPG varies significantly across engine types, with the OHCF engine being the most efficient in mitigating the adverse impacts of higher vehicle weights. To add to these discoveries, fuel type, fuel system, and engine size emerge as significant features in an attempt to predict highway MPG.

Understanding these relationships highlights opportunities for automobile manufacturers to prioritize lighter curb weights, efficient engine designs like OHCF, and fuel systems optimized for higher MPG. By focusing on these factors, the automotive industry can produce vehicles that not only perform better but also reduce emissions, contributing to efforts to combat climate change through improved fuel efficiency and reduced resource consumption. This alignment of performance, efficiency, and sustainability can and will play a crucial role in combating climate change while still meeting consumer needs and wants when choosing a vehicle to purchase.

Throughout this process, one of the main challenges we faced was addressing the non-normal distribution of our data, which significantly impacted our ability to perform standard statistical analyses. Many vehicle attributes, such as highway MPG, curb weight, and engine size, exhibited skewed distributions or heavy tails, which violated assumptions required for parametric tests and linear modeling. To address this, we explored several strategies, including data transformations (e.g., logarithmic or square root transformations) to normalize the distributions. This effort did not end up addressing the issue and we moved forward to testing with this assumption violated.

References

- 1 Schlimmer, Jeffrey. "Automobile." UCI Machine Learning Repository, 1985, <https://doi.org/10.24432/C5B01C>.
- 2 Cheah, Lynette W. (Lynette Wan Ting). "Cars on a Diet : The Material and Energy Impacts of Passenger Vehicle Weight Reduction in the U.S." *Cars on a Diet : The Material and Energy Impacts of Passenger Vehicle Weight Reduction in the U.S.*, Massachusetts Institute of Technology, 1 Jan. 1970, dspace.mit.edu/handle/1721.1/62760.
- 3 Moon, Seoksu. "Potential of Direct-Injection for the Improvement of Homogeneous-Charge Combustion in Spark-Ignition Natural Gas Engines." *Applied Thermal Engineering*, Pergamon, 31 Jan. 2018, www.sciencedirect.com/science/article/pii/S1359431116335931?via%3Dihub.
- 4 "Buying and Driving Clean and Efficient Vehicles | Department of Energy." *US Department of Energy*, www.energy.gov/energysaver/buying-and-driving-clean-and-efficient-vehicles. Accessed 3 Dec. 2024.

Appendix

Reference I. Research Question 1 - Initial Model Summary

```
> initial_model <- lm(data$Price ~ data$City_mpg * data$Highway_mpg)
> View(initial_model)
> summary(initial_model)
```

Call:
lm(formula = data\$Price ~ data\$City_mpg * data\$Highway_mpg)

Residuals:

Min	1Q	Median	3Q	Max
-7966.6	-2076.4	-270.4	899.5	14965.7

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	64866.239	4525.387	14.334	< 2e-16 ***
data\$City_mpg	-1220.477	265.892	-4.590	9.10e-06 ***
data\$Highway_mpg	-1622.636	216.108	-7.508	4.41e-12 ***
data\$City_mpg:data\$Highway_mpg	34.881	4.641	7.515	4.25e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3521 on 155 degrees of freedom
Multiple R-squared: 0.6479, Adjusted R-squared: 0.6411
F-statistic: 95.07 on 3 and 155 DF, p-value: < 2.2e-16

```
> |
```

Reference II. Research Question 1 - Breusch-Pagan test on initial model

```
> bptest(initial_model)
```

studentized Breusch-Pagan test

data: initial_model
BP = 33.429, df = 3, p-value = 2.615e-07

```
>
```

Reference III. Research Question 1 - Shapiro-Wilk test on initial model

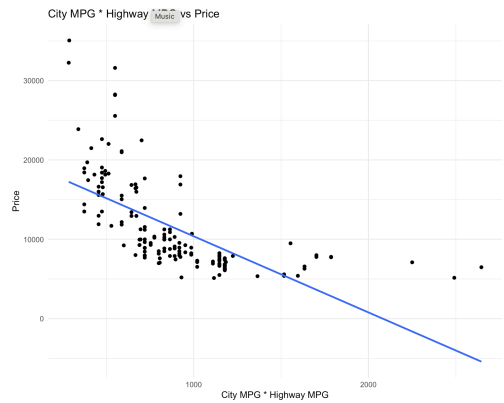
```
> residuals <- residuals(initial_model)
> shapiro.test(residuals)
```

Shapiro-Wilk normality test

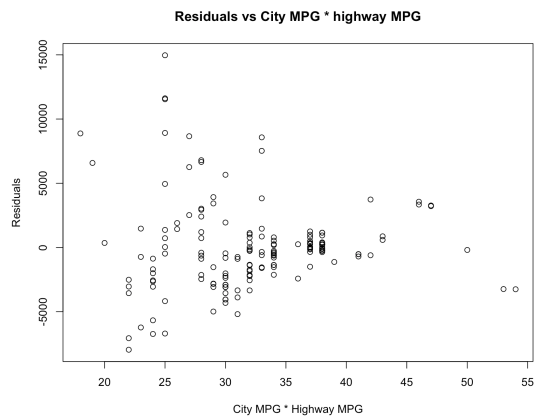
data: residuals
W = 0.90408, p-value = 1.075e-08

```
> |
```

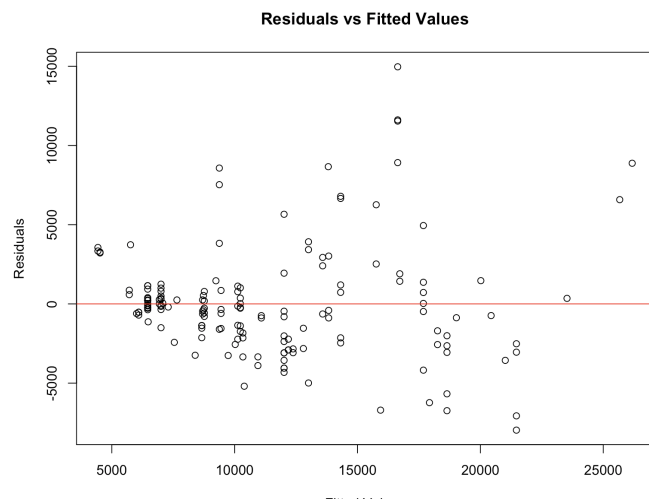
Reference IV, Research Question 1 - Price vs City MPG and Highway MPG plot initial model



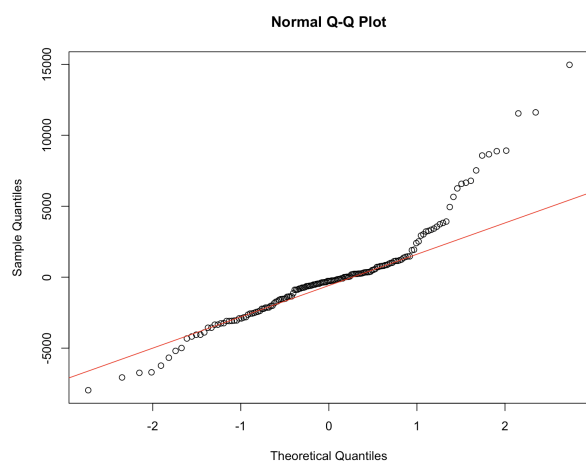
Reference V, Research Question 1 - Residuals vs Predictor Variable Plot initial model



Reference VI. Research Question 1 - Residuals vs Fitted Value Plot initial model



Reference VII. Research Question 1 - Normal QQ plot of initial model



Reference VIII. Research Question 1 - Cook's distance output for initial model

```
> influential_points_cooks
```

```
19 31 48 51 68 69 70 71 73 102 103 126 145 150 204
 9 20 33 34 45 46 47 48 49 72 73 91 104 109 158
```

```
>
```

```
> |
```

Reference IX. Research Question 1 - Breusch-Pagan test on transformed mod

```
> bptest(transformed_model)
```

```
studentized Breusch-Pagan test
```

```
data: transformed_model
```

```
BP = 0.61491, df = 3, p-value = 0.893
```

```
> |
```

```
el
```

Reference X. Research Question 1 - Shapiro-Wilk test on transformed mode

```
> shapiro.test(residuals)
```

```
Shapiro-Wilk normality test
```

```
data: residuals
```

```
W = 0.085164, p-value < 2.2e-16
```

```
>
```

```
> |
```

Reference XI. Research Question 1 - Summary of transformed model

```
> summary(transformed_model)

Call:
lm(formula = data_clean$Price_BoxCox ~ data_clean$Log_CityMPG *
    data_clean$Log_HighwayMPG)

Residuals:
    Min       1Q   Median       3Q      Max
-4.245e-14  2.420e-16  3.260e-16  4.440e-16  5.960e-16

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.701e+00  5.432e-14  3.13e+13 <2e-16 ***
data_clean$Log_CityMPG
-8.926e-15  1.784e-14  -5.00e-01   0.618
data_clean$Log_HighwayMPG
-1.189e-14  1.659e-14  -7.17e-01   0.475
data_clean$Log_CityMPG:data_clean$Log_HighwayMPG
 3.133e-15  4.879e-15   6.42e-01   0.522
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.582e-15 on 142 degrees of freedom
Multiple R-squared:  0.4994,    Adjusted R-squared:  0.4888
F-statistic: 47.22 on 3 and 142 DF,  p-value: < 2.2e-16

>
```

Reference XII. Research Question 2 - Model Summary

```
Call:
lm(formula = horsepower ~ curb_weight + engine_size, data = data,
    weights = weights)

Weighted Residuals:
    Min       1Q   Median       3Q      Max
-5.5621 -2.4507 -0.8589  2.9526  7.8037

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.584619   3.399849  -5.172 7.01e-07 ***
curb_weight   0.016681   0.003949   4.224 4.06e-05 ***
engine_size   0.597774   0.063598   9.399 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.49 on 156 degrees of freedom
Multiple R-squared:  0.9016,    Adjusted R-squared:  0.9003
F-statistic: 714.5 on 2 and 156 DF,  p-value: < 2.2e-16
```

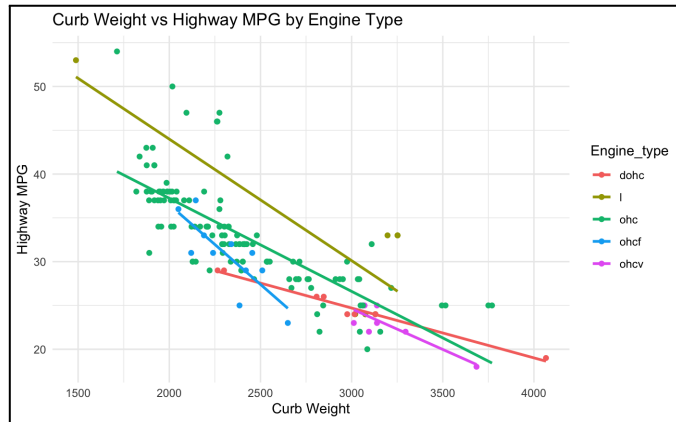
Reference XIII. Research Question 3 - Initial Model Summary & Plot

```
Call:
lm(formula = Highway_mpg ~ Curb_weight * Engine_type, data = data)

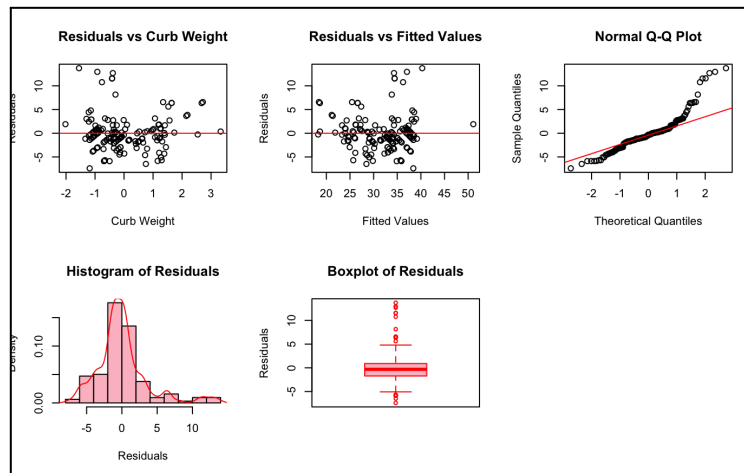
Residuals:
    Min       1Q   Median       3Q      Max
-7.4013 -1.7220 -0.3212  0.9049 13.7179

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   27.7719    1.7747  15.649 < 2e-16 ***
Curb_weight   -2.7434    1.2201  -2.248 0.026015 *
Engine_type1    9.8181    2.4927   3.939 0.000125 ***
Engine_typeohc  4.5602    1.8082   2.522 0.012723 *
Engine_typeohcf 0.3391    2.2765   0.149 0.881796
Engine_typeohcv 1.8339    5.2185   0.351 0.725767
Curb_weight:Engine_type1 -3.9431    1.6815  -2.345 0.020348 *
Curb_weight:Engine_typeohc -2.3779    1.2790  -1.859 0.064977 .
Curb_weight:Engine_typeohcf -6.0347    3.3130  -1.822 0.070536 .
Curb_weight:Engine_typeohcv -1.7248    3.3652  -0.513 0.609038
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

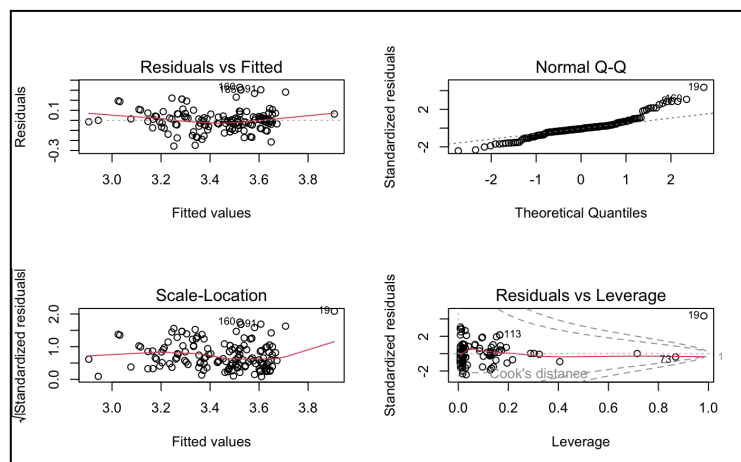
Residual standard error: 3.756 on 149 degrees of freedom
Multiple R-squared:  0.6811,    Adjusted R-squared:  0.6618
F-statistic: 35.36 on 9 and 149 DF,  p-value: < 2.2e-16
```



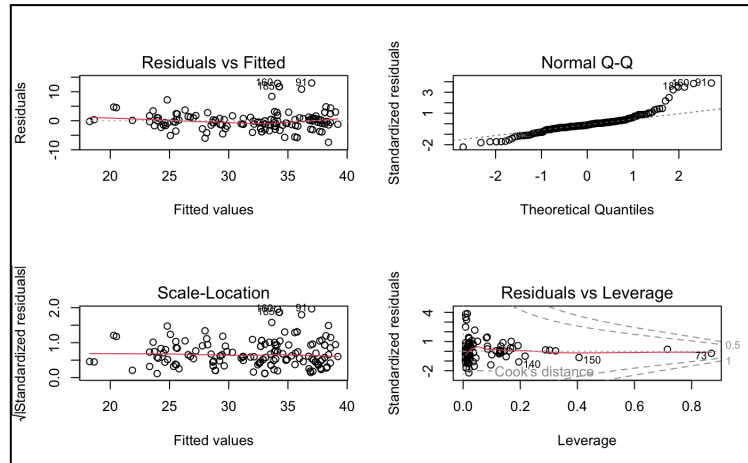
Reference XIV. Research Question 3 - Diagnostic Plots



Reference XV. Research Question 3 - Remedial Log Model Plots



Reference XVI. Research Question 3 - Remedial Outlier Model Plots



Reference XVII. Research Question 3 - Final Log Model Summary

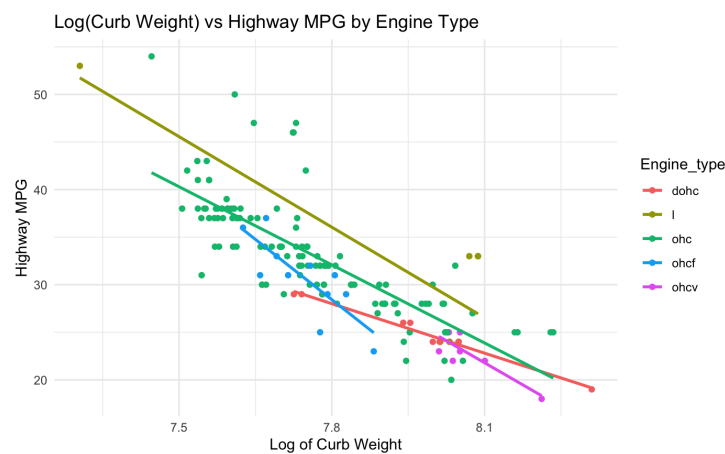
```
Call:
lm(formula = Highway_mpg ~ log(Curb_weight) + Engine_type, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-8.1233 -1.7070 -0.4265  1.1058 12.9852

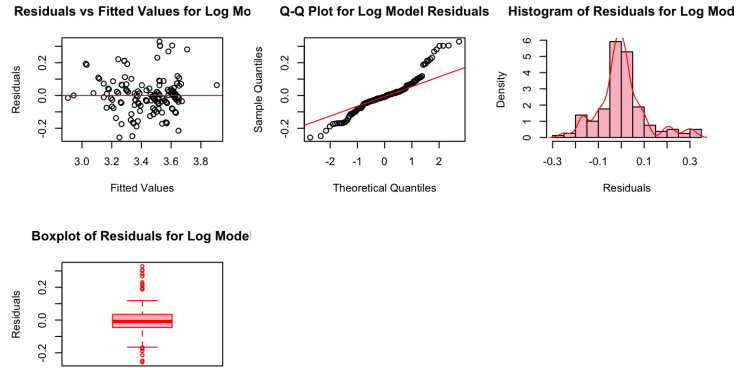
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   244.6213    13.8602   17.649 < 2e-16 ***
log(Curb_weight) -27.5534     1.7325  -15.904 < 2e-16 ***
Engine_type1     5.6799     1.8073    3.143  0.00201 **
Engine_typeohc    2.3738     1.3689    1.734  0.08492 .
Engine_typeohcf  -0.4694     1.6937   -0.277  0.78206
Engine_typeohcv    0.4704     1.8153    0.259  0.79589
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.614 on 153 degrees of freedom
Multiple R-squared:  0.6968, Adjusted R-squared:  0.6869
F-statistic: 70.32 on 5 and 153 DF, p-value: < 2.2e-16
```

Reference XVIII. Research Question 3 - Log(Curb Weight) vs. Highway MPG by Engine Type



Reference XIX. Research Question 3 - Final Model Residual Plots



Reference X: Research Question 4 Model Summary

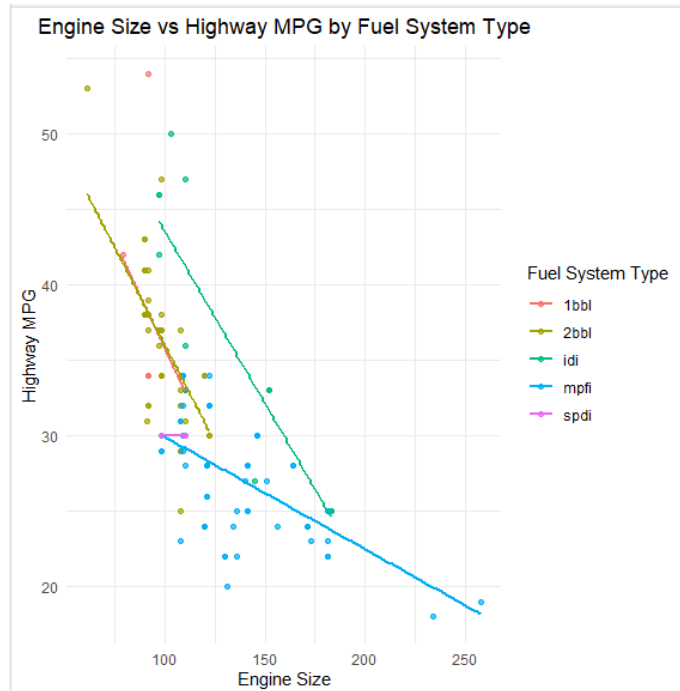
```
Call:
lm(formula = highway_MPG ~ engine_size * factor(fuel_system_combined),
    data = cleaned_data)

Residuals:
    Min       1Q   Median       3Q      Max
-8.938 -1.342  0.000  1.668 15.931

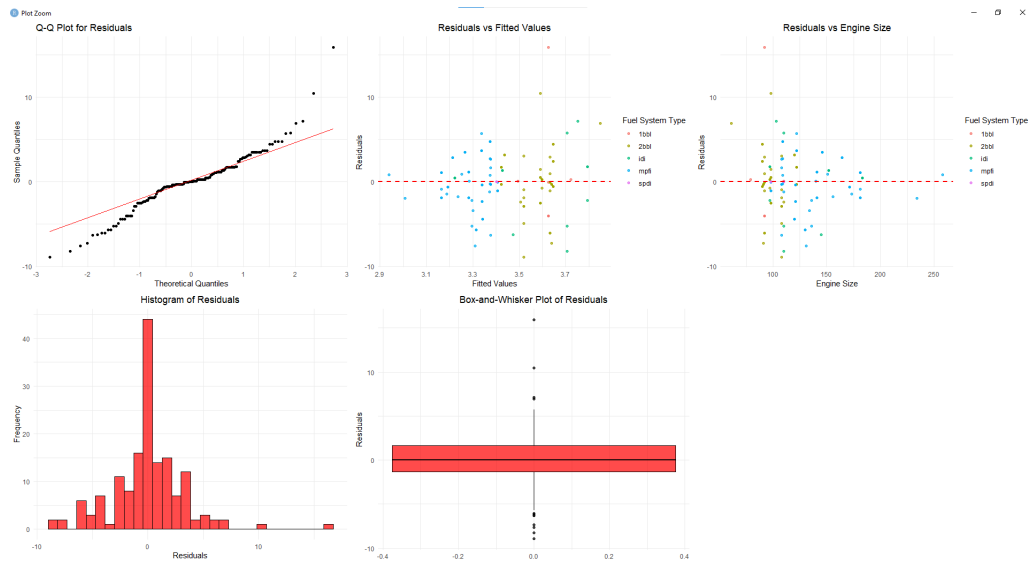
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    64.19630    9.71784   6.606 6.55e-10 ***
engine_size    -0.28400    0.09926  -2.861  0.00483 **
factor(fuel_system_combined)2bb1 -2.37710   10.30771  -0.231  0.81793
factor(fuel_system_combined)idi    2.18663   10.36632   0.211  0.83323
factor(fuel_system_combined)mpfi  -26.80317    9.90578  -2.706  0.00761 **
factor(fuel_system_combined)spdi  -34.19630   28.66201  -1.193  0.23473
engine_size:factor(fuel_system_combined)2bb1  0.02583    0.10492   0.246  0.80584
engine_size:factor(fuel_system_combined)idi   0.05563    0.10250   0.543  0.58811
engine_size:factor(fuel_system_combined)mpfi  0.20943    0.10019   2.090  0.03829 *
engine_size:factor(fuel_system_combined)spdi  0.28400    0.27449   1.035  0.30252
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.364 on 149 degrees of freedom
Multiple R-squared:  0.7442,    Adjusted R-squared:  0.7287
F-statistic: 48.16 on 9 and 149 DF,  p-value: < 2.2e-16
```

Reference XI: Engine Size vs Highway MPG by Fuel System Type



Reference XII: Research Question 4 Diagnostic Plots:



Reference XIII: Research Question 4 Log Model Summary:

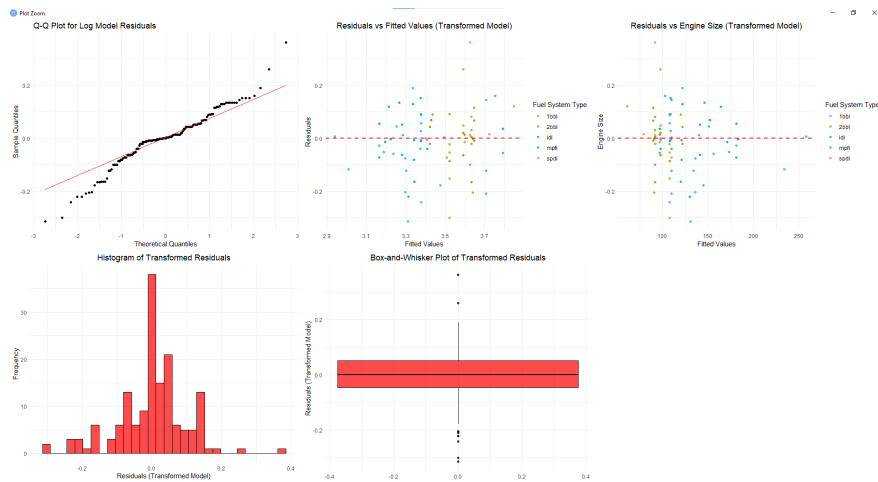
```
call:
lm(formula = highway_MPG_trans ~ engine_size * factor(fuel_system_combined),
  data = cleaned_data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.31450 -0.04565  0.00000  0.05127  0.36256

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.3046626   0.2944039   14.622  <2e-16 ***
engine_size   -0.0073722   0.0030072   -2.451   0.0154 *
factor(fuel_system_combined)2bb1 -0.0283929   0.3122742   -0.091   0.9277
factor(fuel_system_combined)idi    0.1309845   0.3140496    0.417   0.6772
factor(fuel_system_combined)mpfi  -0.6107302   0.3000975   -2.035   0.0436 *
factor(fuel_system_combined)spdi  -0.9034652   0.8683210   -1.040   0.2998
engine_size:factor(fuel_system_combined)2bb1  0.0003724   0.0031785    0.117   0.9069
engine_size:factor(fuel_system_combined)idi    0.0007427   0.0031052    0.239   0.8113
engine_size:factor(fuel_system_combined)mpfi   0.0044431   0.0030352    1.464   0.1453
engine_size:factor(fuel_system_combined)spdi   0.0073722   0.0083158    0.887   0.3768
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1019 on 149 degrees of freedom
Multiple R-squared:  0.7554,    Adjusted R-squared:  0.7406
F-statistic: 51.12 on 9 and 149 DF,  p-value: < 2.2e-16
```

Reference XIV: Research Question 4 Transformed Diagnostic Plots



Reference XV: Research Question 4 Robust Model Summary

```
Call: rlm(formula = highway_MPG_trans ~ engine_size + factor(fuel_system_combined),
  data = cleaned_data)
Residuals:
    Min       1Q   Median       3Q      Max
-0.33587 -0.05818  0.01123  0.04824  0.40689

Coefficients:
              value      Std. Error t value
(Intercept)      3.9611      0.0435   91.1451
engine_size     -0.0041      0.0003  -12.5949
factor(fuel_system_combined)2bb1  0.0272      0.0321    0.8472
factor(fuel_system_combined)idi   0.1117      0.0411    2.7215
factor(fuel_system_combined)mpfi -0.0898      0.0346   -2.5927
factor(fuel_system_combined)spdi -0.1265      0.0530   -2.3887

Residual standard error: 0.08263 on 153 degrees of freedom
```

Reference XVI: Research Question 4 Cross Validation OLS

```
> print(ols_cv)
Linear Regression

159 samples
 2 predictor

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 127, 127, 128, 125, 129
Resampling results:

    RMSE      Rsquared    MAE
3.963736  0.6498282  2.924252
```

Reference XVII: Research Question 4 Cross Validation OLS Log Transformed

```
> cat("Average RMSE (OLS):", round(average_rmse, 3), "\n")
Average RMSE (OLS): 0.111
> cat("Average MAE (OLS):", round(average_mae, 3), "\n")
Average MAE (OLS): 0.085
```

Reference XVIII: Research Question 4 Cross Validation Robust Model

```
> cat("Average RMSE:", round(average_rmse_robust, 3), "\n")
Average RMSE: 0.112
> cat("Average MAE:", round(average_mae_robust, 3), "\n")
Average MAE: 0.084
```